

QUANTITATIVE ANALYSIS: CAUSAL IMPACT OF POLICY INTERVENTIONS

Analytical Framework: A Multi-Level Approach

To rigorously quantify the causal impact of major policy interventions on electric vehicle adoption in India, we employ a hierarchical framework that evaluates effects at both national and state levels. This dual perspective is essential because aggregate national trends may mask substantial geographic heterogeneity in policy effectiveness, while state-level analysis alone may overlook compositional effects that determine overall program success.

We analyze four major interventions: the FAME II subsidy scheme (April 2019), major product launches by Tata and MG (January 2020), the COVID-19 pandemic lockdown (March 2020), and the Production Linked Incentive scheme (September 2021). Our outcome variable is the natural logarithm of monthly EV registrations, which ensures normality, stabilizes variance, and allows coefficients to be interpreted as approximate percentage changes.

National-Level ARIMAX Models

At the national level, we implement Autoregressive Integrated Moving Average with Exogenous Variables models within an Interrupted Time Series Analysis framework (Bernal et al., 2017). ITSA is a quasi-experimental design that compares outcomes before and after an intervention while controlling for pre-existing trends and temporal autocorrelation. The ARIMAX specification decomposes each intervention's effect into two components: an immediate level change capturing any discontinuous jump at policy implementation, and a trend change capturing whether the growth trajectory accelerated or decelerated post-intervention.

Optimal ARIMA orders were selected via the Akaike Information Criterion, testing specifications from ARIMA(0,0,1) to ARIMA(3,0,2). Post-estimation diagnostics confirmed model adequacy, with Durbin-Watson statistics ranging from 1.90 to 2.01 and Ljung-Box tests showing no residual autocorrelation (all $p > 0.05$). Critically, comparison to ordinary least squares revealed that OLS overestimated immediate effects by factors of three to nine due to uncontrolled temporal dependence, underscoring the importance of the ARIMAX approach for time series policy evaluation.

State-Level Panel Analysis

To examine spatial heterogeneity, we estimated panel fixed effects models using state-year data (N=792 observations, 33 states, 24 years). State fixed effects control for time-invariant characteristics such as infrastructure quality, institutional capacity, and baseline adoption propensity, while clustered standard errors account for within-state correlation. This specification estimates the average treatment effect across all states, giving equal weight to each state regardless of size. We tested for heterogeneous treatment effects through interaction terms between interventions and state characteristics, identifying which types of states benefited most from each policy.

State-Level Results and Heterogeneity Patterns

Panel analysis reveals extreme spatial heterogeneity (Table 1). FAME II showed an average effect of positive 30 percent but was statistically insignificant, with state-specific estimates ranging from negative 74 percent in Assam to positive 3,808 percent in Andhra Pradesh. States with near-zero baseline adoption experienced enormous percentage increases from small absolute gains, while established markets showed modest or negative

effects. High-adoption states exhibited 3.2 percentage point weaker responses to FAME II ($p=0.034$), indicating diminishing returns in mature markets.

Product launches showed more uniform impacts with an average effect of positive 19.5 percent ($p=0.01$), but high-adoption states experienced 69 percentage point larger responses ($p=0.04$), suggesting important complementarities between vehicle availability and existing infrastructure. The PLI scheme demonstrated the strongest average effect at positive 53.7 percent ($p<0.001$), but benefits were geographically concentrated in manufacturing hubs. Chhattisgarh showed 327 percent growth, Uttarakhand 270 percent, while some non-manufacturing states experienced negative effects (Table 2). PLI effectiveness declined 2.3 percent per 100 kilometers from manufacturing clusters ($p<0.01$), confirming localized spillovers.

Reconciling National and State-Level Estimates

The substantial divergence between national ARIMAX and panel fixed effects estimates (Table 1) reflects fundamentally different weighting schemes and research questions rather than methodological contradictions. National ARIMAX estimates the marginal effect on aggregate national registrations, weighted by each state's contribution to the total. Large states with high volumes dominate this aggregate. Panel fixed effects estimates the average treatment effect across states, giving equal weight to each state regardless of size. The national estimate answers whether policies achieved aggregate targets; the panel estimate answers whether policies worked for the typical state.

Three mechanisms drive divergence. First, compositional effects mean large states with modest responses dominate national aggregates while small states with large percentage gains have minimal national impact. Second, baseline heterogeneity creates mechanical relationships where states with near-zero baselines show enormous percentage gains from small absolute increases. Third, differential weighting means panel models treat states equally (appropriate for understanding broad effectiveness), while ARIMAX weights by national contribution (appropriate for aggregate assessment).

Consider FAME II. The national estimate of negative 2.9 percent indicates the policy failed nationally, while the panel estimate of positive 30 percent indicates most states experienced growth. The reconciliation: FAME II succeeded in nascent markets like Andhra Pradesh and Jharkhand, generating huge percentage gains but minimal absolute volume. Meanwhile, established markets comprising 70 percent of national registrations showed flat or declining growth. The policy was broadly effective but not nationally impactful.

For PLI, both approaches show positive effects, but panel estimates are larger (53.7 percent versus 10.4 percent). This reflects PLI's concentration in manufacturing states, which are overrepresented in equal-weighted averages relative to their consumption share. Neither estimate is "wrong"—they serve complementary purposes for policy evaluation.

Policy Implications

The multi-level analysis yields several actionable insights. First, supply-side policies outperform demand-side subsidies. PLI showed both immediate and sustained impacts, while FAME II required over a year to manifest effects and proved ineffective in mature markets. Second, one-size-fits-all policies are suboptimal given extreme state heterogeneity. We recommend stage-differentiated interventions: higher subsidies for nascent markets to overcome adoption barriers, infrastructure co-investment requirements for emerging markets, and subsidy phase-outs in mature markets with reallocation to charging networks.

Third, product launches and infrastructure must be coordinated. Product availability was twice as effective in states with existing infrastructure, suggesting that vehicle manufacturers should certify minimum charging availability before launching in new states. Fourth, manufacturing-focused policies require equity adjustments. PLI benefits accrued overwhelmingly to manufacturing hubs, raising concerns about regional divergence. We recommend expanding PLI to component manufacturers in non-manufacturing states and establishing regional investment targets.

Fifth, policy evaluation should routinely incorporate both national and state-level analysis. National estimates inform whether aggregate targets were met; state estimates identify where and why policies worked, guiding optimal targeting and design refinement.

Methodological Contributions and Limitations

Our analysis makes three methodological contributions. First, we demonstrate the importance of controlling for autocorrelation in time series policy analysis, as OLS overestimated effects by factors of three to nine. Second, we illustrate the value of hierarchical frameworks that examine impacts at multiple aggregation levels, revealing complementary insights invisible at either level alone. Third, we advance methods for interpreting divergent estimates, showing how different estimators answer different questions rather than providing contradictory findings.

Several limitations warrant acknowledgment. Temporal confounding remains a concern as product launches, COVID-19, and PLI occurred within 18 months, making complete causal isolation challenging. State-level data quality varies substantially, adding noise to estimates. We identify heterogeneous treatment effects but cannot definitively attribute them to specific mechanisms without additional data on infrastructure, income, and implementation quality. External validity is limited by India's unique context of two-wheeler dominance, dense urban populations, and federal policy structure. Finally, we observe effects of policies as implemented, not under counterfactual optimal designs.